PYTHON FUNDAMENTALS

LISTS

Practice Workbook

From method fluency to multi-step reasoning
50 focused problems. No filler. No answer key.

15	15	15	5
EASY	MEDIUM	HARD	COMPLEX

A focused progression, not a random question dump

This workbook starts with direct list-method practice, then moves into loops, conditionals, duplicates, ordered data, and multi-step problems that require careful reasoning. The questions are intentionally answer-free so the work stays yours.

50	4	0
problems	difficulty levels	solutions included

How to use this workbook

The goal is not to memorize method names. The goal is to predict how a list changes, choose the right operation, and handle edge cases without losing data.

01-15	Easy Direct practice with append, insert, remove, pop, sort, reverse, and extend.
16-30	Medium Methods combined with conditions, loops, duplicate handling, and preserving order.
31-45	Hard Algorithms involving indexes, contiguous patterns, mutation safety, and complexity-aware reasoning.
46-50	Complex Reasoning Small end-to-end systems: command processors, reconciliation, ranking, and data cleanup.

Rules for your solutions

+ Write each solution as a small function unless the question explicitly asks for a script or processor.

+ Test empty lists, one-item lists, duplicates, missing targets, and boundary values where relevant.

+ Notice whether the task asks you to mutate the original list or return a new one. Those are different requirements.

+ For Hard and Complex questions, follow every restriction. A shortcut that violates a constraint is not a valid solution.

> A good habit: before coding, write down the expected output for one normal case and one edge case. Then trace the list after every mutation.

QUESTIONS 01-15

Easy

Build accuracy with the core list methods. Most tasks should take only a few lines.

Working rule: Use direct list operations wherever possible. Pay attention to whether a method changes the list in place or returns a value.

Q01 Build a packing list

Start with an empty list. Add charger, notebook, water bottle, and headphones in that order using `append()`. Print the final list.

Practice focus: append() and insertion order

Q02 Insert a task into second place

Given `tasks = ['check email', 'attend class', 'revise notes']`, insert submit assignment as the second item. Do not rebuild the list manually.

Practice focus: insert() and zero-based indexes

Q03 Remove one ingredient

Given `spices = ['salt', 'cinnamon', 'pepper', 'cumin']`, remove cinnamon from the list and display the updated list. Assume the item appears exactly once.

Practice focus: remove()

Q04 Pop the latest download

Given `downloads = ['report.pdf', 'notes.txt', 'draft.docx']`, remove the last item with `pop()`. Print both the removed filename and the remaining list.

Practice focus: pop() without an index

Q05 Pop a chosen position

Given `days = ['Mon', 'Tue', 'Wed', 'Thu']`, remove the item at index 2 using `pop()`. Display the removed value and the updated list.

Practice focus: pop(index) and index awareness

Q06 Sort exam scores

Sort `scores = [74, 91, 68, 85, 77]` into ascending order using the list's `sort()` method. Print the same list after sorting.

Practice focus: sort() in place

Q07 Rank delivery times

Given `minutes = [42, 35, 51, 28, 39]`, arrange the times from largest to smallest with `sort()`. Do not create a second list.

Practice focus: sort(reverse=True)

Q08 Reverse recently opened files

Given `files = ['home.py', 'lists.py', 'loops.py', 'main.py']`, reverse the list in place and print it.

Practice focus: reverse()

Q09 Combine today's and tomorrow's tasks

Given `today = ['lecture', 'practice']` and `tomorrow = ['revision', 'quiz']`, add every item from `tomorrow` to the end of `today` in one list-method call. Print `today`.

Practice focus: extend() and list growth

Q10 Append a new reading

A log contains `readings = [28, 30, 29]`. Append the new reading 31, then print the updated list and its number of entries.

Practice focus: append() and len()

Q11 Insert a correction

Given `marks = [12, 18, 25, 31]`, insert 99 as the third item. Preserve the order of all existing marks.

Practice focus: insert() at index 2

Q12 Remove only the first matching value

Given `drinks = ['tea', 'coffee', 'tea', 'water']`, remove one occurrence of tea with `remove()`. Print the list and confirm that the other tea remains.

Practice focus: remove() removes the first match

Q13 Serve the next ticket

A queue is stored as `tickets = ['T-104', 'T-105', 'T-106']`. Remove the first ticket using `pop(0)`. Print the ticket served and the remaining queue.

Practice focus: pop(0)

Q14 Alphabetize filenames

Given `names = ['report.py', 'app.py', 'utils.py', 'main.py']`, sort the lowercase filenames alphabetically and print the result.

Practice focus: sort() on strings

Q15 Apply two operations in order

Start with `status = ['queued', 'running', 'waiting']`. Append complete, then reverse the same list. Print the final order. The order of operations matters.

Practice focus: append() followed by reverse()